

The Cruciality of Shor's Algorithm in Breaking RSA Encryption

Bilisuma J. Merga

Abstract

Quantum computing introduces a new computational paradigm that significantly exceeds the capabilities of classical computers for certain problem classes. Classical computers process information using bits that exist in a single state—either 0 or 1—at any given time. In contrast, quantum computers use quantum bits, or qubits, which can exist in a superposition of states, allowing them to represent both 0 and 1 simultaneously. This property enables quantum computers to perform complex calculations in parallel, drastically reducing the time required to solve specific problems that are computationally infeasible for classical systems.

One of the most important applications affected by quantum computing is public-key cryptography, particularly the RSA encryption algorithm. RSA is widely used to secure digital communications and relies on the computational difficulty of factoring very large composite numbers into their prime factors. For classical computers, factoring numbers with key sizes of 512 bits, 1024 bits, or higher requires an impractical amount of time, which has historically made RSA a secure cryptographic system.

The security of RSA is fundamentally challenged by Shor's algorithm, a quantum algorithm introduced by Peter Shor in 1994. Shor's algorithm enables efficient integer factorization by exploiting quantum superposition and interference, reducing the computational complexity from exponential to polynomial time. When executed on a sufficiently powerful quantum computer, this algorithm can effectively break RSA encryption by revealing its private keys.

Although current quantum computers are limited by hardware constraints, error rates, and qubit scalability, experimental implementations of Shor's algorithm demonstrate its feasibility and long-term threat to classical cryptographic systems. This paper reviews the principles of RSA encryption, outlines the fundamentals of quantum computing, and explains how Shor's algorithm compromises RSA security. The discussion highlights the implications of quantum computing for existing cryptographic infrastructures and underscores the growing importance of developing post-quantum cryptographic solutions.

Key words: public key, private key, modular exponentiation period, coprime numbers, greatest common factor(gcf), least common multiple(lcm), Shor's algorithm, modulus of numbers, Hadmard gate